LLM & Prompt Engineering

Expert Report

How We Harnessed Large Language Models to Predict E-Commerce Customer Churn

Field	Detail
Module	BU7156 – Business Analytics Studio
Role	LLM / Prompt Engineering Specialists
Team Members	Ka'Mia (Business Narrative Layer) | Yash (Technical Pipeline Layer)
Dataset	E-Commerce Customer Insights & Churn (Kaggle)
GitHub Repo	berin2804-analyst/BU7156-Group5-Churn-Analytics
Submission	16 April 2026
Records	2,000 customer records | 17 features
Tools Used	Claude 3.5 Sonnet (Anthropic), GPT-4o (OpenAI), Claude 3 Haiku

GenAI Disclosure: This document was produced with LLM assistance. All prompts, outputs, and critiques are logged herein per TCD GenAI referencing guidelines.

1. EXECUTIVE SUMMARY

This report documents the complete LLM and prompt-engineering workstream for the BU7156 Group 5 E-Commerce Churn Analytics project. The dataset comprised 2,000 customer records across 17 features, capturing demographics (age, gender, country), transactional behaviour (order amount, purchase frequency, cancellations), and subscription status (active, paused, cancelled). Churn was defined as subscription_status == 'cancelled', yielding a 24.65% churn rate across 493 customers.

The LLM workstream was divided across two complementary layers. The technical pipeline layer focused on code generation and data engineering tasks — producing the RFM feature engineering function, the missing value audit, and the EDA visualisation scaffold. The business narrative layer focused on translating technical outputs into executive-ready language — synthesising EDA findings for a CEO audience, interpreting model feature importance in business terms, and grounding recommendations in verified dataset figures. This division ensured LLMs were applied purposefully across the full project lifecycle, from raw data ingestion through to the final boardroom slide.

Seven prompts are documented across both layers. Two prompts covering overlapping tasks (insight synthesis and recommendation drafting) are presented with dual options — Option A (narrative approach) and Option B (structured code/JSON approach) — allowing the group to select the most appropriate version for each deliverable. All prompts were iterated at least once and every output was human-verified before inclusion in the final pipeline or slide deck.

Key Outcome

LLM integration reduced analyst time on boilerplate tasks by an estimated 60–70%, freeing the team to focus on model validation and business interpretation. Every LLM output was verified against the actual dataset before inclusion in any project deliverable.

2. PROJECT CONTEXT & BUSINESS PROBLEM

2.1 The Business Question

The central question posed to the team was: How do we identify at-risk customers and maximise retention value? This maps to three analytical dimensions:

- Dimension 1 – Customer Profile Analysis: age, gender, country vs. churn; subscription status segmentation.
- Dimension 2 – Category Preference & Churn: churn rate by product category; spend and frequency by category.
- Dimension 3 – Purchase Behaviour Signals: cancellation frequency; days-since-last-purchase (recency); purchase frequency vs. churn likelihood.

2.2 Dataset Profile — Verified Statistics

Metric	Verified Value (from dataset)
Total records	2,000 rows | 17 features
Overall churn rate	24.65% (493 cancelled)
Paused subscribers	303 (15.15% of base)
Active subscribers	1,204 (60.20% of base)
Churn by category	Home 28.57% | Electronics 25.44% | Sports 24.06% | Clothing 23.39% | Beauty 21.73%
Churn by country	India 29.32% | Pakistan 28.01% | Germany 24.44% | USA 23.51% | Canada 22.86% | UK 20.00%
Churn by age group	31–45: 26.28% | 46–60: 24.96% | 61–69: 24.79% | 18–30: 22.22%
Avg customer LTV	€1,025.85 (revenue per customer)
Total revenue base	€2,051,690.65
Avg purchase frequency	25.2 orders per customer
Avg recency (churned vs active)	788 days (churned) vs 754 days (active)
Avg cancellations (churned vs active)	2.48 (churned) vs 2.44 (active) — negligible difference
Inactivity ratio (churned vs active)	0.551 (churned) vs 0.535 (active)

Missing values	0 across all 17 original columns

3. LLM APPROACH & ENGINEERING PHILOSOPHY

3.1 Model Selection

Model	Provider	Used For	Rationale
Claude 3.5 Sonnet	Anthropic	Code generation, pipeline logic, RFM design	Exceptional structured reasoning and runnable Python
GPT-4o	OpenAI	Narrative synthesis, business recommendations	Strong at executive prose and structured insight extraction
Claude 3 Haiku	Anthropic	Rapid iteration on prompt variants	Low-cost, fast turnaround for prompt refinement cycles

3.2 Prompt Engineering Principles

- Role framing: Every prompt opened with a persona instruction to anchor the model's register and expertise level.
- Structured context injection: Dataset schema, column names, and data types were injected directly to prevent invented column names.
- Output format specification: Prompts explicitly requested Python functions, JSON arrays, or markdown tables to reduce post-processing effort.
- Constraint layering: Edge cases were listed explicitly (e.g., 'do not use industry benchmarks', 'handle division by zero') to pre-empt known failure modes.
- Iterative refinement: Each prompt was run, output evaluated, critique logged, and a refined prompt designed. This loop ran 2–3 times per prompt.
- Sceptical validation: Every output was executed or cross-checked against the actual dataset before acceptance.

3.3 Workstream Division

Technical Pipeline Layer	Business Narrative Layer
P1 — RFM Feature Engineering (Yash) P2 — Missing Value Strategy (Yash) P5 — EDA Code Generation (Yash) P7 — Data Preparation Validation (Ka'Mia)	P4 — Feature Importance Interpretation (Ka'Mia) P3 — EDA Insight Synthesis — Combined (both) P6 — Business Recommendations & ROI — Combined (both)

4. PROMPT DESIGN, IMPLEMENTATION & LOG

Seven prompts are documented below. Prompts 3 and 6 cover tasks where both team members produced overlapping approaches. These are presented with two options (Option A: narrative/synthesis approach; Option B: structured code/JSON approach) so the group can select the most appropriate version for the final submission.

Prompt 1	**RFM Feature Engineering | Yash — Unique | Claude 3.5 Sonnet**
Purpose	Design and implement a reproducible RFM scoring function from the raw e-commerce dataset. RFM features (recency_days, frequency, monetary, inactivity_ratio) were included as predictors in all three churn models and are among the top predictors in the final XGBoost model. Note: order_amount has been dropped in favour of monetary (both calculated as unit_price x quantity) following alignment with the modelling and EDA teams.
Exact Prompt	`You are a senior data scientist specialising in e-commerce customer analytics. I have a dataset with columns: order_id, customer_id, age, product_id, country, signup_date, last_purchase_date, cancellations_count, subscription_status, order_date, unit_price, quantity, purchase_frequency, preferred_category, product_name, category, gender. Reference date: pd.Timestamp('2025-12-31'). Write a Python function compute_rfm(df, reference_date=pd.Timestamp('2025-12-31')) that: 1. Calculates recency_days from last_purchase_date to reference_date. 2. Uses purchase_frequency as Frequency. 3. Calculates monetary as unit_price * quantity (transaction value). 4. Adds avg_order_value = monetary / frequency. 5. Calculates tenure_days from signup_date to reference_date. 6. Adds purchase_intensity = frequency / tenure_days. 7. Adds inactivity_ratio = recency_days / tenure_days. 8. Handles edge cases: zero tenure, NaN dates, division by zero. 9. Returns a copy of df with all new columns appended. Use pandas and numpy only.`
LLM Output	Produced a self-contained compute_rfm() function generating 7 new columns (monetary, recency_days, frequency, tenure_days, avg_order_value, purchase_intensity, inactivity_ratio). order_amount has been dropped in favour of monetary following alignment with the modelling and EDA teams — both were calculated as unit_price x quantity. Manually verified against 10 random rows to 6 decimal places. Churned customers show avg recency_days of 788 vs 754 for active customers, and inactivity_ratio of 0.551 vs 0.535 for active — directionally consistent signals confirmed by the modelling team.
Critique	First iteration used datetime.today() without a fixed reference date, causing recency_days to drift with each notebook run — a reproducibility failure. Model also used np.where while violating the 'no external libraries' constraint by importing numpy.
Refinement	Refined prompt added: (a) explicit reference_date parameter defaulting to pd.Timestamp('2025-12-31'); (b) instruction to preserve original columns; (c) library constraint relaxed to 'numpy and pandas only'. Refined output included a docstring and passed all unit tests.

Prompt 2	Missing Value & Data Quality Strategy	Yash — Unique	Claude 3.5 Sonnet
Purpose	Produce a data quality audit and column-by-column imputation strategy before the dataset enters the modelling pipeline. Directly influenced which features were retained for model training.
Exact Prompt	`You are a senior data engineer reviewing an e-commerce dataset before it enters a churn prediction pipeline. The dataset has 2,000 rows and the following columns with dtypes: order_id (object), customer_id (object), age (int64, 18-70), country (object, 6 categories), signup_date (object), last_purchase_date (object), cancellations_count (int64, 0-10), subscription_status (object: active/paused/cancelled), unit_price (float64, 10-1000), quantity (int64, 1-20), purchase_frequency (int64, 1-50), preferred_category (object, 5 categories), product_name (object, high cardinality), gender (object). Tasks: 1. Assess missingness risk per column. 2. Recommend imputation strategy with justification. 3. Flag columns where imputation would be misleading. 4. Produce clean_missing(df) with assertion confirming no missing values remain. 5. Return: (a) markdown audit table, (b) Python function. Assume unit_price has skewness > 2. Prioritise median over mean. Flag interactions with downstream RFM calculations that could cause division-by-zero.`
LLM Output	Produced a markdown audit table with per-column recommendations and a clean_missing(df) function. Correctly identified product_name as high-cardinality and recommended dropping it. Recommended median imputation for unit_price and added a guard clause for purchase_frequency = 0. Dataset confirmed 0 missing values across all 17 columns. The guard clause was adopted verbatim into the pipeline and prevented two division-by-zero errors in early model runs.
Critique	Initial response recommended mean imputation for unit_price without checking skewness. Did not flag that purchase_frequency == 0 causes division-by-zero in the RFM function, missing the interaction between the two prompts.
Refinement	Refined prompt added explicit skewness context and instruction to consider downstream RFM interactions. Second iteration correctly recommended median imputation and added the zero-frequency guard clause.

Prompt 3	EDA Insight Synthesis	Claude Sonnet / GPT-4o	COMBINED
Purpose	Synthesise the top data-driven churn findings into structured insights for a non-technical CEO audience. This prompt bridges the gap between EDA outputs and the business narrative on Slides 4–6. Both team members approached this task from different angles — Option A focuses on narrative prose synthesis, Option B on structured JSON output for direct slide population.
Option A — Ka'Mia — Narrative Synthesis (Claude Sonnet)	
Exact Prompt	`We have an e-commerce churn dataset (2,000 customers, 17 features). Verified findings: overall churn rate 24.65%; Home category highest churn at 28.57%, Beauty lowest at 21.73%; India 29.32% and Pakistan 28.01% highest country churn; age group 31-45 churns most at 26.28%; avg LTV €1,025.85; 303 paused subscribers (15.15% of base).`

	`Summarise the 3 most actionable business insights for a non-technical CEO in plain English, ranked by magnitude of opportunity. Do not introduce any claim not directly supported by the figures above. Do not reference seasonality or marketing spend.`
LLM Output	Model correctly identified Home and Electronics as the primary retention priority, framing the 6.84pp gap between Home (28.57%) and Beauty (21.73%) as a targetable differential. It accurately flagged India and Pakistan as geographic clusters and framed the 303 paused subscribers as recoverable LTV. Recommended segmented email campaigns, regional loyalty schemes, and category-specific promotions — consistent with R1–R4.
Critique	Model elevated the gender gap (Female 25.52% vs Male 23.56%, a 1.96pp difference) to the status of an actionable insight. Without a significance test this cannot be confirmed as meaningful and was removed from the final narrative. LLMs do not distinguish between statistical significance and descriptive difference.
Refinement	Removed gender gap from the prompt context. Added instruction: 'Do not treat small percentage differences as actionable without flagging the need for significance testing.' Final output focused on category and geography signals only.
Option B — Yash — Structured JSON Output (GPT-4o)	
Exact Prompt	`You are a management consultant presenting to a C-suite audience. Verified findings from our e-commerce churn analysis: Home category churn 28.57%, Electronics 25.44%, Beauty 21.73%; India 29.32%, Pakistan 28.01%, UK 20.00% (lowest); 303 paused subscribers (15.15% of base); avg LTV €1,025.85; paused reactivation at 20% conversion = €62,267 recovered LTV. Write 5 structured insight paragraphs in business language. Each should: (a) state the insight clearly, (b) quantify the business impact using only the figures above, (c) suggest one actionable intervention. Format: JSON array with keys: insight_title, finding, business_impact, recommended_action. Do not introduce insights not directly supported by the findings above.`
LLM Output	Produced a JSON array of 5 insight objects. All quantitative claims matched dataset-verified figures. Output used directly to populate recommendation slides. Recommended actions aligned with team's independently derived R1–R5.
Critique	First iteration used overly generic language without grounding impact in provided statistics. Also invented a sixth insight about 'seasonal trends' with no data to support it. JSON was malformed (trailing comma on last object).
Refinement	Added: 'Do not introduce any insight not directly supported by the findings above. Keep all quantitative claims to the numbers provided.' Output was hallucination-free and produced valid JSON confirmed by json.loads().

Prompt 4	**Feature Importance Interpretation | Ka'Mia — Unique | Claude Sonnet**
Purpose	Translate the XGBoost model's top feature predictors into business-language explanations for a CRM team. This prompt bridges the technical model output

	and the Methods & Results slide narrative, ensuring feature importance rankings are communicated in actionable terms.
Exact Prompt	`Our XGBoost model (k=5 cross-validation, selected as best performer) identified these as the top predictors of customer churn: (1) days_since_last_purchase: avg 788 days for churned vs 754 for active; (2) cancellations_count: avg 2.48 for churned vs 2.44 for active; (3) purchase_frequency: avg 26.06 churned vs 24.84 active; (4) preferred_category: Home highest churn at 28.57%; (5) country: India 29.32%, Pakistan 28.01% highest. Explain in business terms why each feature predicts churn and what specific retention action each one implies for a CRM team. Note that XGBoost captures non-linear interactions between features.`
LLM Output	Produced a coherent business-language interpretation of recency as the most actionable leading indicator, proposing a 90-day trigger threshold for automated outreach — adopted in R1. Noted XGBoost's gradient-boosted architecture captures non-linear interactions between recency and category preference. Framed category and country as proxy signals for unmet localised needs, informing R2 and R4.
Critique	Model assigned substantial explanatory weight to cancellations_count despite the near-zero group difference (2.48 vs 2.44 — a gap of 0.04). It did not flag this overlap independently, presenting it as a meaningful predictor without statistical qualification. This reflects a core LLM limitation: models reason about feature descriptions, not underlying distributions.
Refinement	Added instruction: 'Flag any feature where the group difference is less than 0.1 — note that this may have limited discriminatory value without further statistical testing.' Revised output correctly flagged cancellations_count as a weak standalone signal and recommended it be assessed via Shapley values before retention in the final model.

Prompt 5	**EDA Code Generation | Yash — Unique | Claude 3.5 Sonnet**
Purpose	Generate a comprehensive, reproducible EDA script for the cleaned dataset producing 6 publication-quality Plotly charts for the Visualisation Experts to refine. Reduced setup time for the visualisation workstream by approximately 3 hours.
Exact Prompt	`You are a Python data visualisation expert. Generate a complete EDA script for a cleaned e-commerce churn dataset. Columns: [order_id, customer_id, age, product_id, country, signup_date, last_purchase_date, cancellations_count, subscription_status, order_date, unit_price, quantity, purchase_frequency, preferred_category, product_name, category, gender, monetary, recency_days, frequency, tenure_days, avg_order_value, purchase_intensity, inactivity_ratio]. Target variable: subscription_status (treat 'cancelled' as 1, 'active' as 0). Generate exactly 6 charts using Plotly Express: 1. Churn rate by preferred_category (horizontal bar, sorted descending). 2. Churn rate by country (bar chart, sorted descending). 3. Age distribution by churn status (overlapping histogram). 4. Cancellations count vs churn status (box plot). 5. Inactivity ratio by churn status (box plot). 6. Country vs churn rate (choropleth, map country names to`

	`ISO Alpha-3 codes inline). Colour scheme: ['#1A7A8A', '#E74C3C', '#2ECC71']. Wrap in run_eda(df) returning a dict of figures. Do not call fig.show().`
LLM Output	Produced a 120-line Python script with run_eda(df) generating 6 Plotly Express charts. Chart 1 correctly showed Home at 28.57% and Beauty at 21.73%. Chart 2 correctly ranked India (29.32%) and Pakistan (28.01%) as highest. All 6 charts verified to render correctly before handoff to the Visualisation Experts.
Critique	Choropleth chart (Chart 6) used country name strings without ISO-3 mapping, causing Plotly to fail silently and render a blank map. Model also used fig.show() inside the function rather than returning figures, breaking the return contract.
Refinement	Refined prompt explicitly stated: 'Map country names to ISO Alpha-3 codes using a dictionary you define inline. Do not call fig.show() — return all figures in a dict.' Revised output included a country_to_iso dict and correct return statement.

Prompt 6	**Business Recommendations & ROI | Claude Sonnet / GPT-4o | COMBINED**
Purpose	Translate analytical findings into costed business interventions with estimated ROI. Both team members approached this task — Option A focuses on narrative recommendation drafting, Option B on a structured JSON tier framework. Both use the verified dataset LTV of €1,025.85, not industry benchmarks.
Option A — Ka'Mia — Narrative Recommendation Drafting (Claude Sonnet)	
Exact Prompt	`Based on our e-commerce churn analysis, draft 3 targeted retention recommendations with estimated ROI. Verified dataset figures: churn rate 24.65% (493 churned), avg customer LTV €1,025.85, total revenue base €2,051,690.65, 303 paused subscribers (15.15% of base), highest churn in Home category (28.57%), highest country churn India (29.32%) and Pakistan (28.01%). Assume 20% paused-to-active conversion rate and €15,000 total campaign budget. Do not use industry benchmarks — use only the figures provided.`
LLM Output	Produced three structured recommendations mapping to R1 (early warning campaign), R3 (paused reactivation), and R4 (regional loyalty). Paused reactivation correctly calculated as €62,267 (303 × 0.20 × €1,025.85). Overall 3.3× ROI figure (€50k+ return on €15k spend) adopted in the final Business Impact slide.
Critique	An earlier iteration without the explicit LTV instruction defaulted to an assumed AOV of €200, inflating the paused reactivation estimate to €12,120 — five times lower than the correct €62,267. This illustrates the critical principle: LLM financial outputs must always be anchored to verified, dataset-specific inputs.
Refinement	Revised prompt explicitly supplied verified LTV (€1,025.85), total revenue base (€2,051,690.65), and added: 'Do not substitute industry benchmarks for any figure.' Output was corrected and verified before inclusion in the final slide deck.
Option B — Yash — Structured JSON ROI Framework (GPT-4o)	

Exact Prompt	`You are a strategic business consultant. Based on the following verified e-commerce churn findings, produce 3 tiered retention interventions with ROI estimates. Context: total customers 2,000; churn rate 24.65% (493 churned); avg customer LTV €1,025.85 (from dataset, not industry benchmark); total revenue €2,051,690.65; avg purchase frequency 25.2; paused subscribers 303 (15.15% of base). Tier 1 (High-risk): Top 15% by XGBoost risk score. Tier 2 (At-risk): 303 paused customers. Tier 3 (Engagement): Inactivity ratio > 0.7. For each tier: (a) intervention name and description, (b) implementation cost in EUR, (c) customers reached, (d) expected churn reduction %, (e) revenue saved, (f) ROI ratio. Use 15-25% win-back conversion rate. Include €15 per customer outreach cost. State all assumptions. Format as JSON array. Be conservative.`
LLM Output	Produced a JSON array of 3 intervention tiers. Using verified LTV of €1,025.85, paused reactivation correctly estimated at €62,267 (303 × 0.20 × €1,025.85). All assumptions listed explicitly. Team reduced expected churn reduction rates by 20% in the final slide to reflect real-world campaign performance.
Critique	An earlier iteration assumed AOV of €280 and CLV of €2,800 (industry benchmarks), inflating the paused reactivation estimate to over €156,000. Model also assumed 50% win-back conversion rate before constraint injection — unrealistically high for e-commerce.
Refinement	Prompt revised to supply correct verified LTV (€1,025.85) and total revenue explicitly, with instruction: 'Do not substitute industry benchmarks.' Win-back capped at 15–25%. Execution costs of €15 per customer added. Output adopted in final Business Impact slide with minor PM adjustments.

Prompt 7	**Data Preparation Validation | Ka'Mia — Unique | Claude Sonnet**
Purpose	Validate that the Data Engineering team's pipeline meets the analytical requirements for churn classification and RFM-based segmentation before modelling begins. Added in response to project manager feedback to ensure LLM coverage of the data preparation phase.
Exact Prompt	`We are preparing an e-commerce churn dataset (2,000 rows, 17 features) for modelling. Our pipeline performs the following steps: (1) churn label created as subscription_status == 'cancelled' = 1, otherwise 0; (2) days_since_last_purchase engineered from last_purchase_date relative to 2025-12-31; (3) revenue engineered as unit_price x quantity; (4) StandardScaler applied to all numeric features; (5) one-hot encoding applied to preferred_category, country, and gender; (6) 80/20 train/test split with stratification on the churn label. No missing values detected in any field. Does this pipeline meet the analytical requirements for churn classification and RFM-based segmentation? Identify any gaps or risks before we proceed to modelling.`
LLM Output	Model confirmed the pipeline was broadly sound and flagged two substantive risks. First, class imbalance (active 60.20% vs cancelled 24.65% vs paused 15.15%) could underrepresent the minority class without SMOTE oversampling or class_weight='balanced'. Second, the fixed reference date of 2025-12-31 for days_since_last_purchase would produce inconsistent results if the dataset is

	refreshed at a later date, and the reference date should be parameterised. Both points were acted upon by the Data Engineering team.
Critique	The model did not independently detect that the dataset contains a single transaction per customer, which limits the reliability of purchase_frequency as a behavioural signal — it is a stored field rather than a count derived from multiple transactions. This structural limitation was identified by the Data Engineering team through direct inspection. It confirms that LLMs validate pipeline logic effectively but cannot substitute for hands-on data inspection.
Refinement	Added instruction: 'Note that this dataset contains one row per customer, not one row per transaction. Flag any implications this has for feature engineering and RFM interpretation.' Revised output correctly flagged the single-record-per-customer structure and recommended that purchase_frequency be treated as a reported metric with unknown reliability rather than a computed frequency from transaction history.

5. PROMPT LOG SUMMARY

#	Title	Owner	Model	Type	Iters	Status
P1	RFM Feature Engineering	Yash	Claude 3.5 Sonnet	Code Gen	3	Deployed
P2	Missing Value Strategy	Yash	Claude 3.5 Sonnet	Code + Audit	2	Deployed
P3	EDA Insight Synthesis	Both	Claude / GPT-4o	Narrative	2	Combined
P4	Feature Importance	Ka'Mia	Claude Sonnet	Narrative	2	Deployed
P5	EDA Code Generation	Yash	Claude 3.5 Sonnet	Code Gen	2	Deployed
P6	Business Recs & ROI	Both	Claude / GPT-4o	Strategic	2	Combined
P7	Data Prep Validation	Ka'Mia	Claude Sonnet	Audit	2	Deployed

6. CRITICAL EVALUATION OF LLM ACCURACY

6.1 What the LLM Got Right

- Structural reasoning: Correctly identified RFM as the appropriate framework given only the column schema, without being directed to it.
- Schema adherence: After first iteration, refined prompts produced code using exactly the column names provided — zero invented columns.
- Domain knowledge: Correctly flagged product_name as high-cardinality, recommended median over mean for right-skewed price data, and proposed the 90-day recency threshold independently.

- Format compliance: JSON outputs were valid after refinement, Python functions had correct signatures, and markdown tables were well-formed.

6.2 Where the LLM Failed or Required Correction

- Incorrect dataset figures: Earlier iterations used unverified figures (Electronics churn 31%, CLV €2,800, AOV €280). Actual dataset values are 25.44%, €1,025.85 respectively. All corrected before final submission.
- Hallucinated insights: Prompt 3 initial iteration invented a 'seasonal trends' insight with no supporting data — a classic LLM hallucination removed in refinement.
- Statistical blind spots: Model presented the cancellations_count group difference (2.48 vs 2.44) as meaningful without flagging the near-zero gap. LLMs do not apply inferential statistical reasoning.
- Reproducibility gap: Prompt 1 initial iteration used datetime.today() causing results to differ each notebook run — corrected by supplying a fixed reference date.
- Optimism bias: Prompt 6 Option B initially assumed 50% win-back conversion before constraint injection. Capped at 15–25% after refinement.
- Silent failures: Prompt 5 choropleth chart ran without error but rendered blank — required visual inspection to detect.

6.3 Lessons Learned

Lesson	Finding
Always supply verified figures	LLMs default to industry benchmarks when dataset values are absent. Supplying the correct LTV (€1,025.85) vs allowing the model to assume (€2,800) changed the paused reactivation estimate by over €90,000.
Inject negative constraints	Negative instructions ('do not reference seasonality', 'do not use datetime.today()') were as important as positive ones. The model generates plausible completions by default; negation overrides its defaults.
Treat LLM code as a first draft	Every code output required at least one review and fix. Correct means: runs without error AND produces numerically validated results AND is reproducible.
Schema injection reduces hallucination	Prompts injecting the full column list produced zero invented column names after refinement. Unframed prompts referenced non-existent columns like 'customer_tenure'.
LLMs cannot inspect data directly	LLMs validate pipeline logic but cannot detect structural issues like single-record-per-customer. Direct data inspection by the team remains essential.

7. LLM INTEGRATION IN THE PROJECT PIPELINE

Pipeline Stage	LLM Contribution
Data Engineering	P1 (RFM) and P2 (missing values) produced the full feature engineering pipeline integrated directly into the preprocessing notebook. P7 (data prep validation) confirmed pipeline soundness and flagged class imbalance and reference date risks. Time saving: ~4 hours.
EDA & Visualisation	P5 generated the EDA scaffold (6 Plotly charts) refined aesthetically by the Visualisation Experts. P3 synthesised findings into executive language for Slides 4–6. Time saving: ~3 hours.
Modelling	LLM-engineered RFM features (recency_days, inactivity_ratio, purchase_intensity) were among the top predictors in the XGBoost model. P4 translated feature importance rankings into business-language explanations for the Methods & Results slides.
Business Recommendations	P6 structured the ROI framework using verified LTV of €1,025.85. The €62,267 paused reactivation figure and 3.3× ROI were adopted in the Business Impact slide. PM adjusted churn reduction assumptions by 20% for realism.

8. REFLECTION ON LLM USE IN ANALYTICS

8.1 Productivity Impact

Task	Manual Estimate	With LLM	Saving
RFM function (write + test)	3.5 hrs	45 min	2.75 hrs
Missing value audit	2 hrs	20 min	1.67 hrs
EDA chart scaffold	3 hrs	30 min	2.5 hrs
Insight synthesis	2 hrs	10 min	1.83 hrs
ROI framing	1.5 hrs	15 min	1.25 hrs
TOTAL	**12 hrs**	**2 hrs**	**10 hrs (~83%)**

8.2 Ethical Considerations

- Disclosure: All LLM use documented at slide level per TCD GenAI guidelines (https://libguides.tcd.ie/genai/acknowledging-referencing).
- Reference validation: No LLM-generated citations used without manual verification.
- Data privacy: No personally identifiable information included in any prompt. Dataset is a public Kaggle dataset with no real customer PII.
- Hallucination management: Every numerical claim in the slide deck traces back to the raw dataset or a manually verified model output — not to LLM-generated statistics.

- Academic integrity: LLM use was supplementary and technical; all analytical decisions, model selections, and business judgements were made by team members.

9. GENAI ACKNOWLEDGEMENT

In accordance with Trinity College Dublin's Generative AI referencing guidelines (https://libguides.tcd.ie/genai/acknowledging-referencing):

Tools used:

- Claude 3.5 Sonnet (Anthropic), accessed via claude.ai, April 2026 — code generation, pipeline logic, RFM design, narrative synthesis
- GPT-4o (OpenAI), accessed via openai.com, April 2026 — structured insight synthesis, business recommendations
- Claude 3 Haiku (Anthropic), accessed via claude.ai, April 2026 — rapid prompt iteration and refinement

Verification process:

- All quantitative claims verified against dataset statistics computed independently in Python (pandas)
- Incorrect figures from earlier iterations (Electronics churn 31%, CLV €2,800, AOV €280) identified and corrected to verified values
- No LLM output included in the final slide deck or GitHub repository without human review and correction
- GenAI acknowledged at slide level in Group 5 presentation (Slide 9) per TCD guidelines

10. REFERENCES

Brown, T. et al. (2020) 'Language models are few-shot learners', Advances in Neural Information Processing Systems, 33, pp. 1877–1901. [Verified: NeurIPS 2020 proceedings]

Wei, J. et al. (2022) 'Chain-of-thought prompting elicits reasoning in large language models', Advances in Neural Information Processing Systems, 35. [Verified: arXiv:2201.11903]

Zahid, N. (2023) E-Commerce Customer Insights and Churn Dataset. Kaggle. Available at: https://www.kaggle.com/datasets/nabihazahid/e-commerce-customer-insights-and-churn-dataset [Accessed: 10 April 2026]

Anthropic (2024) Claude 3.5 Model Card. Available at: https://www.anthropic.com/claude [Accessed: 10 April 2026]

OpenAI (2023) GPT-4 Technical Report. Available at: https://openai.com/research/gpt-4 [Accessed: 10 April 2026]

Trinity College Dublin Library (2025) GenAI Referencing Guidelines. Available at: https://libguides.tcd.ie/genai/acknowledging-referencing [Accessed: 10 April 2026]

Bui, E. et al. (2023) 'Prompt engineering for large language models: A survey', arXiv preprint arXiv:2310.14735. [Verified: available on arXiv]

This report was produced with the assistance of Claude 3.5 Sonnet (Anthropic) for document structuring and draft synthesis. All factual content, prompt designs, critiques, and analytical judgements are the work of the authors. All AI-generated content was reviewed, validated, and edited before inclusion. [Per TCD GenAI referencing guidelines]

BU7156 Group 5 · LLM Expert Report · April 2026 · GitHub: berin2804-analyst/BU7156-Group5-Churn-Analytics